ANSI/IES RP-27.1-22

Illuminating
ENGINEERING SOCIETY

RECOMMENDED PRACTICE: RISK GROUP CLASSIFICATION AND MINIMIZATION OF PHOTOBIOLOGICAL HAZARDS FROM ULTRAVIOLET LAMPS AND LAMP SYSTEMS

AN AMERICAN NATIONAL STANDARD

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has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared by the
IES Photobiology Committee

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1.0 Introduction and Scope

1.1 Introduction

The purpose of this Recommended Practice is to summarize the photobiological hazards of exposure to ultraviolet (UV) radiation and to provide recommendations to minimize the risks of such effects from ultraviolet lamps and lamp systems.

Although most lamps and lamp systems (see **Sections 3.18** and **3.19**) are safe and pose no photobiological risks except under unusual exposure conditions, one group of products—ultraviolet lamp systems—may pose optical radiation hazards during use that require risk assessment for direct and indirect exposure of the eyes and skin. Optical radiation hazards from all types of lamps or other broadband light sources are assessed by application of *ANSI/IES RP-27-20, Lighting Science: Photobiological Safety of Lamps and Lamp Systems*[1] (combined from the previously separate ANSI/IES RP-27.1-15, ANSI/IES RP-27.2-00(R2017), and ANSI/IES RP-27.3-17). It addresses LEDs, incandescent, low- and high-pressure gas-discharge, arc, and other lamps. ANSI/IES RP-27.1-22 expands this information by covering lamp systems that are designed primarily to emit ultraviolet radiant energy, such as ultraviolet sources intended to excite fluorescence of irradiated materials, as used in insect light traps, scientific studies, mineral identification, non-destructive testing, germicidal irradiation, and other purposes.

This document provides risk group classification for all ultraviolet lamp systems, and the measurement conditions for different applications, thus superseding the more general risk group classifications used in ANSI/IES RP-27-20 for UV lamp systems. It includes manufacturing and user safety requirements that may be required as a result of an ultraviolet lamp system being assigned to a particular risk group. The assigned risk group of an ultraviolet lamp system also may be used to assist with any needed risk assessments, e.g., for occupational exposure in workplaces.

1.2 Scope

Recommendations in this document apply only to lamps and lamp systems designed primarily to emit ultraviolet (UV) radiant energy for consumer, industrial, scientific, and medical applications. The scope is limited to lamps and lamp systems where more than half of the optical radiation emitted between 180 nm and 3,000 nm is in the spectral region 180 nm to 400 nm. If more than half of the optical radiation emitted between 180 nm to 3,000 nm is outside of the spectral region 180 nm to 400 nm, then the base standard, ANSI/IES RP-27-20, applies.

ANSI/IES RP-27.1-22 does not apply to general lighting sources (GLS) or laser systems.

2.0 Normative References

2.1 ANSI/IES LS-1-22

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2022.

2.2 ANSI/IES RP-27-20

Illuminating Engineering Society. Recommended Practice: Photobiological Safety for Lighting Systems. New York: IES; 2020.

2.3 ANSI/IES RP-44-21

Illuminating Engineering Society. Recommended Practice: Ultraviolet Germicidal Irradiation (UVGI). New York: IES; 2021.

3.0 Definitions

Standard nomenclature and definitions, radiometric and photometric quantities, and illuminating engineering terminology are found in *ANSI/IES LS-1-22, Lighting Science: Nomenclature and Definitions for Illuminating Engineering* (see **Section 2.1**). Other definitions found in ANSI/IES RP-27-20 (see **Section 2.2**) may also apply.

3.1 actinic radiation hazard

Optical radiation, most notably UV-B and UV-C, capable of producing a photobiological effect. The relative spectral hazard is defined by the weighting function *S*(λ), published by ACGIH.

3.2 assessment distance
The distance used to determine the risk group classification of a lamp or lamp system.

Note: Because photobiological effects from UV radiation are based on the total accumulated exposure (dose) received, this standard relies on the concept of time-weighted average exposures, where the assessment distance for determining the risk group is chosen based on realistic exposure distances and exposure durations. In other words, it is not expected that people will be exposed at very close distances—e.g., 20 to 30 cm—for extended periods. This standard provides assessment distances and specific guidance that are application-specific and realistic rather than more-general values where the specific application is unknown and time-weighted average exposures are not application-specific.

3.3 blue light hazard (BLH)
The potential for a photochemically induced retinal injury resulting from radiation exposure at wavelengths primarily between 400 nm and 500 nm. This damage mechanism dominates over thermal mechanisms for intense visible light and for times exceeding 10 seconds but is rarely of concern from UV lamp systems (unless the basic lamp is an arc lamp).

3.4 blue light hazard radiance, L_B
Effective photobiological radiance with the spectral radiance, L_λ, spectrally weighted with the blue light hazard spectral weighting function, *B*(λ). [Source for definition: CIE ILV,[2] item 17-26-057.]

Note 1 to entry: The wavelength range for the blue light hazard radiance is specified in ICNIRP-recommendations as λ = 300 nm to λ = 700 nm.

Note 2 to entry: This definition is based on the assumption that an action spectrum is adopted for the actinic effect considered, and that its maximum value is 1.

Note 3 to entry: It is essential to specify which actinic action spectrum is used, as the unit is the same for any action spectrum.

Note 4 to entry: The blue light hazard radiance is expressed in watt per square meter per steradian ($W \cdot m^{-2} \cdot sr^{-1}$).

Note 5 to entry: This entry was numbered 17-99 in CIE S 017:2011.

3.5 competent person
One who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to people, and who has authorization to take prompt corrective measures to eliminate them. [Source: OSHA 1926.32(f).[3]]

3.6 consumer
A person who purchases goods and services for personal use. Consumers include not only users of an ultraviolet lamp or lamp system but also all persons who may have access to the lamp or lamp system, or who may be near the product. Sometimes referred to as an "ordinary person."

3.7 controlled access location
The location where an engineering and/or administrative control measure is established to restrict access except to authorized personnel with appropriate safety training.

3.8 dose-limited product
A lamp or lamp system where the emitted radiant exposure (dose) is limited by time or actual exposure monitoring at the assessment distance to a set level during any day. The dose limit is expressed in millijoules per square centimeter (mJ/cm^2).

3.9 effective exposure distance
The nearest distance of time-weighted human exposure consistent with the intended use of the lamp or lamp system for which the time-weighted emission has been considered.

3.10 emission limit
A limit defined for each risk group, based upon reasonably foreseeable conditions of exposure. It incorporates exposure duration, exposure distance, and time-weighted averaging, and is derived from exposure limits.

Note: In the rare situation where the BLH is evaluated, the emission limit may be expressed as radiance.

3.11 Exempt Group

The risk group classification for which there is no photobiological hazard.

3.12 exposure limit

The maximum level of exposure to the eye or skin that is not expected to result in adverse biological effects.

Note: For the most current limits, refer to the ACGIH's guide to TLVs,[4] which is updated annually.

3.13 general lighting source lamp (GLS)

As a light source description for risk group classification (of lamps): A general term for a continuous wave (CW) lamp intended for lighting spaces that are typically occupied or viewed by people. Used to differentiate from non-general lighting sources in the process of determining risk group classification.

3.14 germicidal UV (GUV) lamp system

Any UV lamp system designed to disinfect using a UV-C anti-microbial effect to inactivate pathogens so that they are no longer capable of replicating and causing adverse health effects.

3.15 instructed person

A person who has been instructed and trained by a skilled, competent person, or who is supervised by a competent person, to identify ultraviolet sources that may cause adverse effects and to take precautions to avoid unintentional exposure to those sources. [Adapted from IEC 62368-1:2018.[5]]

3.16 intended use

Usage of a lamp, lamp system, process, or service in accordance with specifications, instructions, and information provided by the manufacturer or supplier.

3.17 lamp

A generic term for a manufactured source created to produce optical radiation.

Note: For the purposes of this document, the term includes any kind of electrical source emitting in the optical range, including LEDs and excimer lamp sources.

3.18 lamp system

Any manufactured product or assemblage of components that incorporates, or is intended to incorporate, one or more lamps.

Note: A UV lamp system is one that is intended to emit UV radiation for its primary application, or for which the emission of UV radiation is a major function.

3.19 lamp packaging

Any lamp carton, outer wrapping, or other means of containment that is intended for the storage, shipment, or display of one or more lamps, or that is intended to identify the contents or to recommend its use.

3.20 luminaire, UV

For the purpose of this document: A lamp system that distributes, filters, or transforms the UV radiant energy transmitted from at least one source of optical radiation and which includes all the parts necessary for attaching (sometimes referred to as "fixing") and protecting the sources and, where necessary, circuit auxiliaries together with the means for connecting them to the power supply. UV luminaires commonly are employed in upper-room germicidal applications.

3.21 photocuring lamp system

A lamp system that usually employs UV-A to photopolymerize liquid polymers to a solid state. Examples include UV photopolymerization of liquid inks in printing, and rapid curing of plastic products.

3.22 photokeratoconjunctivitis

A delayed inflammatory response of the cornea and conjunctiva following exposure to UV radiation.

Wavelengths shorter than 320 nm are most effective in causing photokeratoconjunctivitis. The peak of the action spectrum is at approximately 270 nm.

3.23 risk group (RG)

A classification scheme for lamps and lamp systems that indicates the photobiological hazard. Four risk groups, from Exempt to High Risk, are defined. (See **Section 5.4**.)

3.24 time-weighted average (TWA) exposure

The averaged cumulative exposure (dose) delivered over a given period (occurring on a daily 8-hour basis) divided by the exposure duration, to provide an effective irradiance for both variable distances and durations.

Note: The TWA is essential in considering lengthy exposures to UV hazards, since variable exposure distances at different irradiances and durations determine the reasonably foreseeable worst-case exposures (for photochemical hazards), which therefore correspond to the measured or calculated effective irradiance at a specified distance for RG determination (analogous to the 500-lx assessment distance for GLS lamps, as explained in ANSI/IES RP-27-20; see **Section 2.2**).

3.25 ultraviolet radiation

Radiant energy within the wavelength range 10 nm to 400 nm.[6]

Note: For *photobiological* purposes, the subdivisions in the UV range between 100 nm and 400 nm are[6]:

- UV-A: 315 to 400 nm
- UV-B: 280 to 315 nm
- UV-C: 100 to 280 nm

3.26 UV fluorescence illuminator

Any UV lamp system designed to illuminate and excite fluorescent materials to permit increased visualization. Examples include "black light" fluorescent illuminators, security-code readers, and Wood's lamp systems for medical applications.

3.27 view-related risk

The risk for intended viewers of a source under application-specific realistic conditions exceeding 1,000 seconds in one day.

4.0 Background

Ultraviolet lamps and lamp systems are employed in a variety of applications, from novelty "black light" illuminators, theatrical spotlights, and fluorescence applications to medical lamps, insect attractants and germicidal irradiation. Typical lamp types are:

- Low-pressure mercury lamps (tubular and compact lamps with UV-emitting phosphors)
- Low-pressure mercury UV-C lamps
- UV LEDs (bandwidth generally 50 nm or less)
- Excimer lamps, e.g., krypton chloride
- UV-filtered metal-vapor discharge lamps
- Filtered UV xenon lamps (pulsed or continuous)
- UV-filtered tungsten-halogen lamps

Lamps are employed most typically in the applications described in **Sections 4.1** through **4.6**.

4.1 Near-UV (UV-A) "Black Light" Sources to View Fluorescent Pigments

Handheld and fixed (i.e., attached) UV-A fluorescence illuminators with typical peak emissions at 350 to 400 nm are used in a variety of applications, including for exciting fluorescent coatings, paints, and pigments, and to visualize the visible fluorescent emission for non-destructive testing. The viewer may or may not wear UV-blocking eyewear, to improve the contrast of the visible florescence emission spectrum or to protect against accidental direct exposure.

Handheld UV-A spotlights or "flashlights" are used in conjunction with fluorescent dyes in non-destructive testing, e.g., of coatings and cracks. Similar sources are used to detect authentication codes on money, bank notes, identification cards, and identification stamps. In all of these applications, the risk group determination is based upon the highest time-weighted exposure assessed at the reasonably foreseeable worst-case exposure condition. In actual use, the UV-A beam is directed away from the viewer and the user would not receive a potentially hazardous exposure.

Permanently installed "black light" systems are used in, for example, the entertainment industry, museums, bars, banks (for currency checking), and bowling alleys, and are installed to encourage viewing of the lit scene rather than the source, other than unintentional or momentary views.

4.2 Near-UV (UV-A) Insect Attractant Lamps and Lamp Systems

Both fixed (i.e., attached) and portable UV-A lamp systems with typical peak emissions generally within the spectral range of 350 to 370 nm are used in a variety of applications for insect light traps (ILTs). Effective installation requires that the ILTs be placed away from individuals so that insects are attracted to the UV-A source and not attracted to individuals or permanently occupied locations. Therefore, the highest time-weighted exposure is assessed at distances of 1 meter or more, even for indoor units, depending upon application.

4.3 Germicidal Ultraviolet (GUV) Lamps and Lamp Systems

GUV lamp systems for ultraviolet germicidal irradiation (UVGI) traditionally use low-pressure mercury-discharge lamps with primary emission at 254 nm (in the UV-C range).[7] Lamp envelopes were originally made from quartz, but today most lamps use UV-C transmitting glass that will block the 185-nm mercury line that produces significant ozone (see **Section 6.4 Ozone**). Radiation from short-wavelength (less than approximately 240 nm) GUV lamp systems can result in increased ozone in the indoor environment. When these systems are used in occupied areas, the lamp's safety commissioning process should include an assessment of potential room occupant exposure to ozone. For germicidal efficacy, the UV lamps themselves are almost by necessity in Risk Group 3.

Traditional GUV lamp systems have generally been designed to block emission of UV into occupied spaces. GUV is employed in water-purification and air-handling systems, and in domestic air-purifiers that generally have interlocked enclosures to prevent potentially hazardous UV exposure during normal operation. Interlocks also normally preclude access to UV even during maintenance or servicing. However, in the control of airborne infectious agents (bioaerosols) such as the tuberculosis bacillus, and measles and influenza viruses, upper-room GUV luminaires are used in some hospitals, prisons, and institutions, where infection control is critical. The aim is to irradiate only the upper-room volume, placing the GUV luminaires at least 2.1 meters (7 ft) above the floor level. Since the system will be Risk Group 3 when measured correctly on axis, proper installation of GUV luminaires ensures that TWA exposure does not exceed the Threshold Limit Value® (TLV®) in the occupied lower part of the room, less than 180 cm (about 5 ft 11 in.) above floor level (refer to *ANSI/IES RP-44-21, Recommended Practice: Ultraviolet Germicidal Irradiation (UVGI)* for additional information; see **Section 2.3**). Acceptable exposure levels above 180 cm will vary depending on ceiling height.

UV LED lamps and lamp systems have been emerging in recent years. Those using UV LEDs for germicidal applications typically emit radiation with peak emission between 265 and 280 nm. The efficiency of germicidal UV LEDs is improving rapidly, including the more recent development of shorter wavelength devices with peak emissions between 220 and 240 nm.

UV-C emission at 222 nm from krypton chloride excimer (KrCl*)[†] lamps has been recently introduced in germicidal lamp systems that have the potential of being used in occupied spaces with lower skin and eye risk to the occupants.[8] The potential risk due to ozone generation may need to be considered, particularly for high-power excimer and unfiltered UV luminaires (see **Section 6.4**).

Note: The Harvard School of Public Health performed assessments at an eye height of 172 cm (5 ft 8 in.)—e.g., for a 180-cm (6-ft) standing human—and at an eye height of 112 cm (3 ft 8 in.), representing a reclining person in the highest hospital bed, as measured from the floor. Today, 180 cm, corresponding to the 99th-percentile height of the eyes of a standing person, is used.

Additional information on GUV lamps and lamp systems may be found in *IES CR-2-20-V1, IES Committee Report: Germicidal Ultraviolet (GUV) – Frequently Asked Questions*[9] and in *ANSI/IES RP-44-21, Recommended Practice: Ultraviolet Germicidal Irradiation (UVGI)* (see **Section 2.3**).

4.4 UV Tanning Equipment

Tanning devices are not covered in this standard. However, such devices shall comply with *FDA 21CFR*

[†] The symbol for the krypton chloride excimer is KrCl*; in this symbol, the "*" indicates the excited state of the excimer molecule.

1040.20, Performance Standard for Sunlamp Products[10] or equivalent regulation outside the United States. This FDA regulation provides all optical radiation performance requirements designed to minimize risks to the user. Hence, this Recommended Practice does not provide additional requirements or emission limits.

4.5 UV Nail Curing and Treatment

UV lamps and lamp systems for photocuring of nail polish and nail laminates employ UV-A lamps installed in a fixture that is designed to minimize ocular exposure. Irradiances of approximately 10 to 20 mW·cm^{-2} are typical at the nail.[11] Two assessment-distance criteria are required: 1) for the skin at 1 cm; and 2) for the eyes at 20 to 50 cm, since this provides a conservative TWA viewing distance, considering the usage time of a professional operator. Normally, these lamps and lamp systems should be in RG-0 or RG-1. Cautionary wording would be applicable for RG-2 (see **Section 7.4**).

4.6 UV Medical and Dental Sources

Medical lamps and lamp systems using a UV-A ("black light") fluorescent or filtered incandescent lamp are widely used in diagnostic instruments. Frequently referred to as "Wood's lamps," these lamps are found in emergency rooms, ophthalmology departments, and dermatology departments for fluorescence detection and fluorimetry. UV-A-induced fluorescence of proteins and DNA is useful in a range of applications. Dental examinations employ UV-A radiation for identification of tooth plaque, and it has been used for photocuring in restorative dentistry. Medical-laboratory UV trans-illuminators (both UV-A and UV-B) are used to view DNA or RNA that has been separated by electrophoresis through an agarose gel, which are visible by their fluorescence. Assessment distance varies by application. UV-A and UV-B radiation is used in a variety of medical treatments (e.g., dermatology phototherapy).

5.0 Emission Limits and Measurement for Risk Group Determination

5.1 Emission Limits

Table 5-1 provides emission limits (expressed as irradiance) for UV lamps and lamp systems. Normally, UV lamps and lamp systems will not emit visible or IR-A radiation (780 to 3,000 nm) that would approach retinal thermal, blue light, or low-luminance retinal thermal limits, as provided in ANSI/IES RP-27-20 (see **Section 2.2**). A complete list of emission limits for UV lamps and lamp systems is provided in ANSI/IES RP-27-20.

5.2 Time-Weighted Averaging (TWA) of Exposures

TWA is a critical aspect of assessing the risks from UV lamp systems. It is recognized that cumulative daily exposure over variable exposure distances with different irradiances and exposure durations exist for almost all UV applications. The assessment distance (see **Section 5.3**) is determined for the reasonably foreseeable worst-case TWA exposure and is equivalent to that distance where the photochemical hazard would exist after considering the normal movements of the individual in relation to the source during the day (see **Tables 5-2a** through **5-2c,** in **Section 5.4**). Time-weighted averaging is therefore a method to obtain an exposure dose integrated over longer periods during which the irradiance may change. **Annex B** provides additional information on TWA.

Table 5-1. Emission Limits for Risk Groups of Ultraviolet Lamps and Lamp Systems

Risk[a]	Metric	Formula in RP-27-20	Exempt, RG-0 (max 8 h)	RG-1 (max 2.8 h)	RG-2 (max 1,000 s)	Units
Actinic UV, S(λ)	E_s	§ 5.4.1	0.1	0.3	3.0	μW·cm^{-2} (effective)[b]
Near UV, 315 – 400 nm	E_{UV}	§ 5.4.2	1.0	3.3	10	mW·cm^{-2}

Table notes:

a. Although highly unlikely, a UV lamp or lamp system with significant short-wavelength visible radiant emission could require assessment for blue-light-hazard spectrally weighted radiance (L_B). If concern exists, refer to ANSI/IES RP-27-20 for limits (see **Section 2.2**).

b. Spectrally weighted by $S(\lambda)$.

5.3 Measurement and Assessment Distances for UV Lamp Systems

Where possible, measurements of effective UV-B and UV-C irradiance and/or spectral irradiance from UV lamp systems normally should be made at the relevant risk group assessment distance given in **Tables 5-2a** through **5-2c** (see **Section 5.4**). It is often the case, however, that the source UV irradiance is low, so it is desirable to measure the irradiance sufficiently close to the source, i.e., less than the risk group assessment distance, to minimize measurement errors from detector dark current (noise). This is particularly true if high resolution spectral measurements are to be obtained. The measured irradiance shall then be transferred to the relevant risk group assessment distance given in **Tables 5-2a** through **5-2c** (see **Section 5.4**). The measurements should be transferred to the assessment distance through a known relationship (ratio) between the measured and assessed distances. This ratio may be experimentally determined or known from historical mathematical methods.

When the assessment distance for a UV luminaire is significantly less than 3 times the largest dimension of the source, the inverse-square law is not applicable because the distance is too close. In such a case, the ratio shall be experimentally determined. For example, if a 1-m linear fluorescent-tube luminaire is measured at 20 cm normal to and at the center of luminaire output, and the assessment distance is 50 cm in **Table 5-2c**, then the transfer ratio for the irradiance at 50 cm needs to be determined experimentally. A good way to obtain the transfer ratio in this case is to use a broadband detector and obtain the irradiance ratio at both the measured and assessed distances. This experimentally determined ratio is then applied to the spectral measurements. Using the inverse-square law in this case would have led to large errors in the assumed emission at the assessment distance.

However, in the unlikely case that the radiant output is too high and may damage or saturate the instrument detector, the measurement distance will be greater than the assessment distance. In this case, the distances from the source may be sufficiently large to obtain the transfer ratio using the inverse-square law.

For a broadband UV-A source, it may be necessary to determine the stray UV-B irradiance. In this case, spectral measurements obtained at a short measurement distance, e.g., 20 cm to improve signal-to-noise ratio, will be used to determine the UV-B irradiance. The UV-B irradiance at the assessment distance, e.g., 50 cm, can then be obtained by using a broadband irradiance detector to determine the ratio needed to find the UV-B irradiance at 50 cm.

In all cases, the actual measurement values or the extrapolated measurement values at the assessment distance should be reported.

5.4 Risk Group Assessment Distance Criteria

From **Section 5.3**, the irradiance due to the source is now known at the assessment distance. This information, along with the TWA exposure values, permits the evaluation of the risk group using the values shown in **Table 5-1**.

Tables 5-2a through **5-2c** provide the risk group assessment distance criteria for specific UV lamp systems. **Table 5-2a** applies to lamps and lamp systems intended for use by *consumers in uncontrolled environments* and which require no restrictions other than the provision of user information. Under normal or abnormal operating conditions, ordinary persons should not be exposed to UV sources at dose levels capable of causing injury.

Lamps and lamp systems requiring special use training and generally only administrative controls are listed in **Table 5-2b**; these are normally intended for *professional use or use by instructed persons*. Under normal operating conditions, abnormal operating conditions, or single-fault conditions, instructed persons should not be exposed to parts comprising UV sources at dose levels capable of causing injury. Lamps and lamp systems intended for use by consumers (i.e., ordinary persons) or instructed persons shall not emit RG-3 emission levels.

Those lamps and lamp systems requiring engineering controls and/or personal protective equipment as well as administrative controls and special training, and typically limited to professional use by *competent persons*, are listed in **Table 5-2c**.

Table 5-2a. Risk Group Assessment Distances for Consumer (Unrestricted-Use)[a] UV Lamps and Lamp Systems

Lamp System	Assessment Distance (cm)
Fluorescence-detection illuminators, handheld	100
Fluorescence “black light” room and staging illuminators	200
Fluorimeters, transilluminators	50
Home-use skin-clearing illuminators	50, but see **Section 4.5**
Insect light traps (<50 W) and domestic use	100
Nail-curing devices, home-use	50; also, limited to the dose limit at manufacturer’s use distance.[b]
Surface cleaning, handheld (see **Section 3.3**)	20
Water-disinfection systems	100

Table notes:

a. For use by consumers, i.e., persons without special training or skills.

b. The value at this distance can be obtained via a calculation, using either the inverse-square law or the 1/distance relationship, depending on whether the source approximates a point source or a linear source (see **Section 5.3**).

Table 5-2b. Risk Group Assessment Distances for Professional Restricted-Use UV Lamps or Lamp Systems Intended to Be Used by Instructed Persons

Lamp System	Assessment Distance (cm)
Dental diagnostic illuminator	100; but see **Section 4.6**
Dental photocuring illuminator	50; apply emission dose limit at manufacturer’s use distance
Medical diagnostic	Use emission dose limit at manufacturer’s use distance
Medical treatment	See IEC 60601-2-57
Ophthalmic diagnostic devices	See ISO 15004-2
Nail-curing device, professional	20 (assessment for ocular hazards); see **Section 4.5**[a]
Insect light traps: 50 to 150 W: >150 W:	 200 300
Photocuring devices (e.g., inks)	100
Germicidal in-duct irradiation and air-purifier units	50 (from any access point)
Fluorescence testing	60
Lacquer hardening or curing	100
Powder curing	100
Optical-fiber curing	100
Surface activation	100
Sunlamp devices or indoor tanning appliances	See FDA Performance Standard for Sunlamp Products: 21 CFR 1040.20
Surface cleaning, handheld	60
Germicidal upper-room and whole-room irradiation systems	180 (above intended floor level)
Germicidal air purifiers (leakage radiation from the enclosure)	40

Table note:

a. The value at this distance can be obtained via a calculation, using either the inverse-square law or the 1/distance relationship, depending on whether the source approximates a point source or a linear source (see **Section 5.3**).

Table 5-2c. Risk Group Assessment Distances for UV Lamps or Lamp Systems Intended for Use by Professional, Competent Persons

Lamp System	Assessment Distance (cm)
Dental photocuring illuminator	50; but see **Section 4.6**, at contact
Medical treatment	See IEC 60601-2-57
Whole-room germicidal systems for occupied spaces[a]	100, on axis
Whole-room germicidal systems for Restricted Access spaces	50
Upper-room germicidal systems[a]	50 and 100
Germicidal, in-duct system	50 (from access point)
Water-purification systems	50 (from access point)

Table note:

a. Additional safety requirements up to 180 cm above the intended floor level; see ANSI/IES RP-44-21.

For UV lamps and lamp systems not listed in **Tables 5-2a** through **5-2c**, the assessment distance shall be 1 m, or the manufacturer's recommended-use distance in the case that lengthy exposure times (i.e., greater than 100 s) at this distance are feasible.

Some lamp systems are designed to be dose-limited, in which case the risk group classification may be performed by comparing the emission limit with the value calculated by dividing the maximum possible dose emitted by the maximum possible exposure time in an 8-hour period (e.g., 0.1 $\mu W \cdot cm^{-2}$ = 3 $mJ \cdot cm^{-2}$/30,000 s.)

After determining the RG at the appropriate assessment distances given in **Tables 5-2a** through **5-2c,** there is an additional requirement for lamps and lamp systems intended to expose the skin or eyes. The UV-A irradiance at the manufacturer's specified use distance from the tissue surface should not exceed 100 $mW \cdot cm^{-2}$ at the tissue surface in order to preclude overheating.

6.0 Systems Safety Control Measures for Manufacturers

6.1 General Requirements

Systems safety control measures should be applied to lamp systems employing Risk Group 3 (RG-3) lamps; lamp systems using lamps in RG-0, RG-1, and RG-2 are exempt from such control measures. The most commonly encountered UV RG-3 luminaires are for germicidal applications. The design and construction of the final products should limit the access to UV dose levels that could exceed the applicable actinic UV hazard exposure limit for the skin and eye at the assessment distances listed in **Section 5.3**. In almost all cases, the effective UV-A radiation [$S(\lambda)$-weighted] can be eliminated as a risk, and therefore the concern is with UV-B and UV-C radiation.

Where relevant product standards provide specific requirements, these may be applied.

Note: Product safety standards can have additional or more-restrictive requirements, or not permit specific construction or function details.

Conformity is checked by inspection or by commissioning the system after installation.

6.1.1 Germicidal Lamps and Lamp Systems. Lamps, lamp systems, and products containing lamps that emit UV-C for germicidal purposes shall limit access by humans to UV-C emission levels that could exceed the applicable actinic UV hazard exposure limit for the skin and eye (i.e., hazardous UV-C). There are two broad categories of UV luminaires, one that will permit adjustment of the beam alignment at the time of installation by a competent installer, and one that is not designed to allow such adjustments. Installation of these general types are covered in **Sections 6.1.2** and **6.1.3**, respectively.

6.1.2 Upper-Room Germicidal UV Luminaires, Allowing Beam Alignment. Adjustable luminaires that maximize upper-room germicidal activity, while maintaining safety in the occupied space are generally preferred by competent installers. UV luminaires that allow beam alignment fall into four categories:

a. Fixed luminaire with adjustable baffles (i.e., louvers)
b. Fixed luminaire with an adjustable reflector
c. Luminaire with adjustable mounting position
d. Any combination of the above

It is important to understand that for option "a" or "b" above, these types of adjustment will also affect the total output of the luminaire. To aid the installer, some manufacturers also allow the adjustment of power to the luminaire, to help ensure safety.

For manufacturer quality-assurance tests, it is important that the measurements be reproducible and be made in a manner that will reflect the intended use in the field. The tests should clearly indicate to the designer and the installer the irradiance on a horizontal reference plane at specified distances away from the fixture. The horizontal reference plane shall be 1.8 m above the finished floor (AFF).

The importance of adjustable luminaires is that they may be adjusted downward or upward to maximize the irradiance in the upper room and minimize the irradiance in the occupied space. Therefore, an adjustment range of 5° to 10° for the luminaire, the baffles, or the reflector is highly recommended. Any product quality-assurance test should be performed with the baffles, reflector, and luminaire in the neutral position as defined by the manufacturer. (See **Figure 6-1**.)

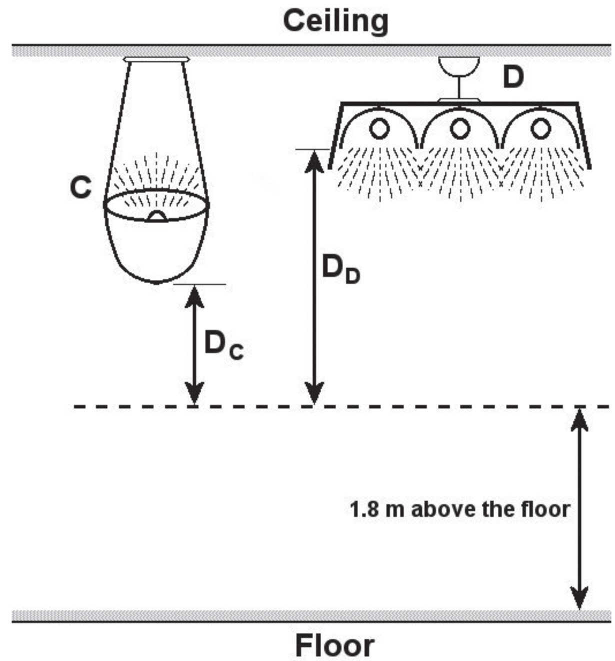


Figure 6-1. Test condition alignment for non-baffled adjustable-position luminaires. "D_C" and "D_D" indicate the height above 1.8 m to luminaire C and luminaire D, respectively. (Graphic courtesy of David Sliney)

6.1.3 Upper-Room Germicidal UV Luminaires, With Fixed-Beam Alignment. Upper-room UVGI luminaires that do not have a means of adjusting the UV beam permit the manufacturer to specify a mounting height AFF such that UV irradiance levels in the occupied space, below 1.8 m AFF are safe. However, it is important that the manufacturer provide clear installation instructions that include verification by the installer that there are no objects in line with the emitted beam that could reflect UV radiation above exposure limits into the occupied space. While most surfaces do not reflect well in the UV-C region,[12] this is not true for all materials. The specification and the user manual of such luminaire shall provide detailed instructions on how to ensure compliance. (See **Figure 6-2**.) Manufacturer testing should require an upper-room UV luminaire to be in a fixed position with a 1°-downward aiming direction.

6.1.4 Upper-Room Indirect UV Luminaires. Because this type of luminaire is typically open (no louvers; see, for example, luminaire "C" in **Figure 6-1**), it may be used in rooms with ceilings higher than 4 m. Manufacturers shall provide instructions for proper installation of the particular UV luminaire. This type of luminaire requires installation by a competent person.

6.1.5 Direct Downward Luminaires. This type of luminaire may be used in rooms with ceilings higher than 3 m. Manufacturers shall provide instructions for proper installation of the particular UV luminaire. This type of luminaire requires installation by a competent person.

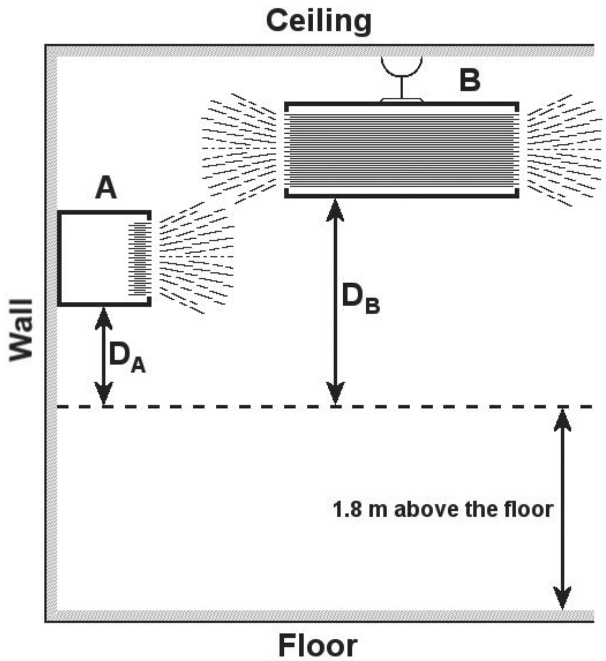


Figure 6-2. Test condition alignment for luminaires with non-adjustable baffles. "D_A" and "D_B" indicate the height above 1.8 m to luminaire C and luminaire D, respectively. (Graphic courtesy of David Sliney)

SIDEBAR: Time-Weighted Averaging and Manufacturers' Guidelines

Realistic, reasonably foreseeable exposure conditions are to be reflected in any manufacturer's testing. The important photobiological risk assessment depends upon the time-weighted average (TWA) exposure. An assumption of lengthy, 8-hour continuous exposures in a standing position, facing the upper-room UV luminaire, may be unreasonable for some lamps or lamp systems.[i] The ACGIH document describes TWA exposure and how to properly interpret TLV exposure limits. The use of these guidelines or standards by the installer should be recommended in the user's manual.

i. American Conference of Governmental Industrial Hygienists. (2022). 2022. TLVs and BEIs: Based on the Documentation of the Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. Ultraviolet Radiation. Cincinnati, OH: American Conference of Governmental Industrial Hygienists.

6.2 Protective Housing

The UV lamp or lamp system shall have a protective housing that, when in place, prevents access to potentially hazardous UV-B and UV-C radiation not required for the intended function of the product.

Note: A protective housing can act alone and is then only effective when it is in place. A protective housing can also act in conjunction with an interlock or proximity sensor; in this case, protection is ensured regardless of the position of the protective housing.

6.2.1 General. Enclosures used as protective housing should follow all of the following:

- Have the necessary rigidity to withstand reasonably foreseeable mechanical impacts during normal use
- Be made from a material not degraded by UV irradiation in a way that could increase the accessible hazardous actinic UV emissions
- Be constructed such that access to hazardous actinic UV is only possible with the use of a tool

Conformity is checked by inspection and appropriate tests, e.g., an impact test.

6.2.2 Openings, Panels, and Doors. Openings, panels, and doors that allow access to hazardous UV-B and/or UV-C radiation should conform with one of the following:

- Be constructed such that the use of a tool is required to open any lids, panels, or doors
- For any service or maintenance openings intended for instructed persons, be provided with a non-defeatable interlock system
- For any service or maintenance openings intended for competent persons, be provided with either of the following:
  - a non-defeatable interlock system
  - a defeatable interlock system, and an emission warning system or warning device; the means to defeat the interlock system may be a password, physical key, or other specific tool

6.2.3 Exemptions Applicable to RG-2 and RG-3 Products. Where the intended use requires access to potentially hazardous UV-B and UV-C radiation, the product shall be provided with alternative means for ensuring protection against hazardous UV-B and UV-C radiation. This may include the mandate to wear specific personal protection devices. The requirements shall be part of the product and training manual.

6.3 Requirements to Aid in Safe Handling and Maintenance

6.3.1 Proximity Sensor. Where exposure to potentially hazardous UV-B or UV-C radiation is possible, a proximity sensor or other means of detecting human presence shall be used in lieu of a protective housing or an interlock system. The proximity sensor shall automatically terminate the emission within 1 second in the event of encroachment by personnel into the area of primary emission (for example, where RG-2 irradiation limits would be exceeded). The proximity sensor shall prevent lamp turn-on if presence of an individual is detected prior to delayed turn-on of the system. If the technology or application would preclude this termination of emission, the manufacturer shall provide a risk assessment as to whether the emission (measured as radiance or irradiance) is reduced below the limit of RG-2 at distances from the source specified by **Table 5.2c**.

Conformity is checked by inspection or by commissioning the system after installation.

6.3.2 Delayed-On Timer. For products intended to be used in whole rooms or upper-room spaces, a delayed-on switch, to allow users to move to a safe distance prior to emission, may be used in lieu of an interlock device or proximity sensor.

Conformity is checked by inspection during commissioning of the system after installation.

6.3.3 Exposure Time Control and/or Auto-shutoff. For products intended to be used in whole rooms or upper-room spaces (see **Table 5-2c**), an automatic shut-off timer that will disable emissions after a specific amount of time, to allow users to access the product after intended exposure, may be used in lieu of an interlock or proximity sensor.

Conformity is checked by inspection during commissioning of the system after installation.

6.3.4 Emission Warning. Where emission exceeding RG-1 levels can occur (or where interlocks can be defeated), the product shall have an audible or visible warning. The warning shall give a visible or audible signal when the system is switched on. Lamps (such as low-pressure mercury) that emit visible radiation during UV emission shall be considered as meeting the visible emission warning.

Conformity is checked by inspection during commissioning of the system after installation.

6.3.5 Emission Controls. The product shall be provided with controls allowing the operator safe handling of the equipment.

6.3.5.1 Emissions Stop. A permanently attached means to disable hazardous UV-B and UV-C emissions shall be provided.

Conformity is checked by inspection during commissioning of the system after installation.

6.3.5.2 Key Control. Products intended for use by competent persons only, and where it is reasonably foreseeable that these may be left unattended in a domestic or commercial setting, shall be provided with a key-operated or electronically based master control.

This key shall be captive in the "on" position, and any hazardous UV-B and UV-C emissions shall not be possible when the key is removed. This may be accomplished via a password, physical key or other specific tool.

Note: Products left unattended in a domestic or commercial setting that can be activated by a consumer or an ordinary person can inadvertently cause a potentially hazardous situation.

Conformity is checked by inspection during commissioning of the system after installation.

6.3.6 Orientation Control for Handheld Products. Handheld consumer products shall have an orientation control that disables the hazardous UV-B and UV-C emission when the product is moved out of normal use orientation.

Manufacturers of handheld products intended for professional users (see **Table 5-2c**) shall provide adequate instructional materials, training and, if needed, personal protection guidance.

Orientation control may be via a tilt sensor or photoelectric switch.

Orientation control may not be an acceptable means of risk reduction, or sufficient if used alone, for products intended to be used in a domestic setting, or for products intended for use in a commercial setting if the products may be left unattended.

Conformity is checked by inspection.

6.4 Ozone

Products that have a potential of generating ozone[13] (emitting UV below approximately 240 nm, unless labeled as ozone-free), should be evaluated at the installation site unless they have been tested to applicable standards.

Depending on the room size, the number of air exchanges per hour, and the number of units installed, the potential for exceeding ozone concentration limits should be evaluated.

If the potential for ozone emission exists, instructions and warnings for monitoring or control of ozone level near the product shall be provided in the manufacturer-supplied instructions.

Where concern exists, the concentration of ozone may be checked with an ozone meter. The instrument should be capable of measuring as low as 0.05 ppm.

Note: UL provides two special-products standards related to this topic: UL 867[14] and UL 2998.[15] OSHA provides permissible-exposure limits,[16] as does NIOSH.[17]

7.0 Labeling

The lamp or lamp system should be permanently marked with the required labels, as described in the subsections that follow.

7.1 User Information

The manufacturer shall provide the user sufficient information to operate the lamp or lamp system safely depending on the risk group (RG) of the lamp or lamp system. The user information shall include the risk group classification of the lamp or lamp system and any required precautions. If the lamp or lamp system is classified as RG-3, the user information shall indicate that this system is "For Professional Use Only" or "For Installation by Professionals Only," as applicable. If fixture options (e.g., tilting mechanisms or modifying optics) can be used with the system, possible worst-case exposure hazard values and hazard distances shall be listed in the user manual.

Note: In the wavelength range 200 nm to 400 nm, where the risk-group classification is based on irradiance or radiant exposure, it should be noted that the *assessment of a lamp cannot be automatically transferred to the final lamp system.*

User information is particularly important in germicidal applications. ANSI/IES RP-44-21 (refer to **Section 2.3**) provides safety information.

SIDEBAR – Labeling Requirements

It is important to note that only the appropriate label for the lamp or lamp system are required, and contradictory labeling should be avoided. Experience shows that multiple, conflicting warnings – even though intended for different readers – can produce a negative reaction by the user that does not serve the interest of safety.

7.2 Warning Signs

Warning signs should be posted at the access points for all areas where potentially hazardous UV emissions are present. If practical, temporary warning signs are preferred because they draw greater attention. All warning signs should be written in the languages relevant to the state, province, country, or user community. Warning signs should include a signal word (e.g., Caution, or Warning), a brief description of the hazard, and instructions to mitigate the hazard. A sample warning sign is provided in (see **Figure 7-1**).

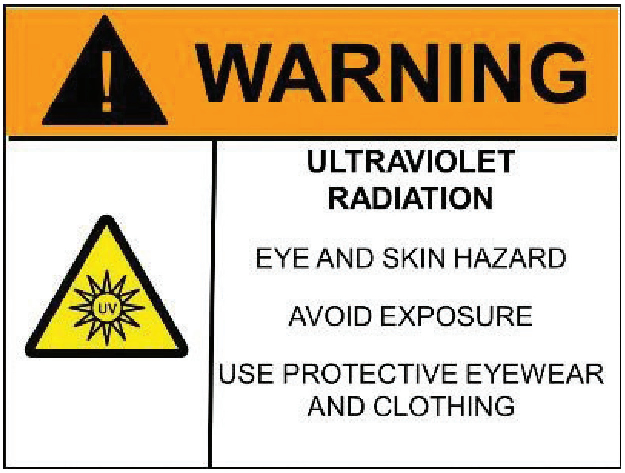


Figure 7-1. Sample UV warning sign with three recommended components: (a) signal word (Warning); (b) identification of hazard (UV radiation; eye and skin hazard); and (c) instructions to mitigate the hazard (avoid exposure; use protective eyewear and clothing). Designs and specific messaging will vary depending on lamp system hazards and control measures used.

7.3 Labeling for UV Lamps, Packaging, and Instructions

Required labeling for UV *lamps* is described below:

RG-0 UV lamps:
No label is needed.

RG-1 UV lamps:
User instructions and lamp packaging shall state that UV radiation is emitted from the lamp.

RG-2 UV lamps:
User instructions and lamp packaging shall include a label stating: "UV radiation is emitted from this product." If any special-use or safety instructions are required for safe use, the ISO M002 symbol, meaning "Refer to instruction manual/booklet," shall be used (see **Figure 7-2**).

Figure 7-2. The ISO symbol used to refer the user to the instruction manual or booklet.

RG-3 UV lamps:
A warning label shall be applied to the user instructions and lamp packaging as specified in IEC 60417[18] or as shown in **Figure 7-3.** The label shall use this (or equivalent) wording: "WARNING: UV radiation is emitted from this lamp. Skin or eye injury could result. Avoid exposure of eye and skin to unshielded lamp."

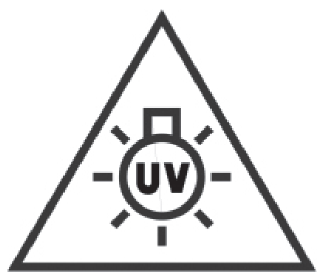


Figure 7-3. Graphic 6040 of UV lamp inside triangle from IEC 60417.[18]

7.4 Labeling and User Information for UV Lamp Systems

For UV lamp *systems*, one of the optional informative labels shown in **Figure 7-4** may be applied as appropriate. The symbols are based on the symbol used in IEC 61549[19] for UV-C germicidal lamps and shall be used for all UV lamp systems. Normal intended use, relamping, and installation requirements shall be described in detail in the user manual. If required, training and restricted-area requirements shall be specified in information for the user.

In addition, risk group labeling shall be applied as applicable, as described below:

RG-0 UV lamp systems:
UV lamp systems that are classified as RG-0 require no additional labeling. However, a label indicating that UV radiation is emitted is optional.

Figure 7-4. Optional labels to provide added information for systems with narrow-band UV lamps.

RG-1 UV lamp systems:
A label indicating that UV radiation is emitted is optional. However, there should be information provided in the user instructions indicating that UV radiation (and the specific part of the UV waveband, e.g., UV-C, UV-B, or UV-A, or the emitted peak wavelength in the case of LEDs) is emitted from this product.

RG-2 UV lamp systems:
The assigned risk group and the hazard shall be labeled in languages as required by the relevant national regulatory body.

There should be information provided in the user instructions indicating that UV radiation (and the specific part of the UV waveband, e.g., UV-C, UV-B, or UV-A, or the emitted peak wavelength in the case of LEDs) is emitted from this product.

Instruction manuals and product information shall state: "Avoid direct exposure to eyes or skin." A label (IEC 61549) shall be applied on the product similar to one shown in **Figure 7-3** or **7-4** as appropriate.

RG-3 UV lamp systems:
The assigned risk group and the hazard shall be labeled in languages as required by the relevant national regulatory bodies. There should be information provided in a label on the system and in the user instructions indicating that UV radiation (and the specific part of the UV waveband, e.g., UV-C, UV-B, or UV-A, or the emitted peak wavelength in the case of LEDs) is emitted from this product.

This warning from IEC 60417 shall be used: "WARNING: Turn off the UV lamp before opening" and "WARNING: Use UV radiation eye and skin protection during servicing."

Instruction manuals and product information shall state: "No direct exposure to the UV radiation from this product shall be allowed."

For UV lamps with removable modifying optics, each modifying optic shall carry a label indicating whether the hazard is increased with the use of the optics.

For germicidal UV lamp systems: A label shall be applied on the UV lamp system similar to: "Warning! Allow no direct exposure of the eyes or skin to the UV radiation from this product."

For vitamin D lamps and phototherapy lamps: A label shall be applied on the UV lamp system similar to: "Warning: Allow no direct exposure of the eyes to UV radiation."

8.0 Safety Control Measures for Occupied Spaces

Professional-user control measures are necessary because RG-3 UV systems can be installed or used inappropriately—particularly in occupied spaces. All UV lamps and lamp systems should be installed and used in such a way as to minimize needless, inappropriate human exposure to UV radiation. This section discusses general control for products intended to be installed or used by professional persons. Whole-room and upper-room UVGI provide the primary example of widespread use of RG-3 UV lamps and lamp systems. Upper-room UVGI is used to disinfect the air in the upper room above an occupied space. By contrast, whole-room UVGI irradiates the entire room, but means exist to preclude direct human exposure above applicable limits by limiting emission into occupied areas by design or by proximity sensors and similar means. This section reviews the lamps and lamp systems that have the potential for human exposure if not properly installed and provides information for post-installation testing.

ANSI/IES RP-44-21 (see **Section 2.3**) provides additional information on comparative applications.

SIDEBAR – Safety and Efficacy Information for the User

The wavelength(s) and irradiance at 1 m from the UV luminaire should be specified to aid the user in estimating efficacy for a given bioaerosol. The efficacy of some UVGI lamps and lamp systems has been highly criticized by public health experts. The public health concern has been that buyers and installers might read that a lamp or lamp system is certified to meet IEC safety requirements, but that system could also be quite ineffective. It should be clearly specified that a safety certification does not imply effectiveness. For example, if a buyer of a "germicidal UV product" (lamp or lamp system) were to purchase a UV-A luminaire with a one-log disinfection efficacy per day, they might think it equivalent to a UV-C lamp or lamp system that provides a one-log disinfection every 10 minutes. It is important that both safety and effectiveness be made clear.

8.1 Post-installation Testing of Upper-Room RG-3 UV-C Systems

Safe installation requires a system that prevents potentially hazardous stray radiation in the lower room, so as not to place occupants needlessly at risk. Experience over more than half a century demonstrates that accidental overexposures of room occupants occur in almost all cases as a result of incorrect installation without effective final onsite safety evaluation. UV luminaires that are well designed can be readily and properly installed to prevent these hazardous situations. Typically, the installer is responsible for the final commissioning process (final acceptance testing) once the UV luminaires are positioned as required based upon the room layout and air circulation patterns. Thus, requirements and testing of a manufacturer's UVGI lamp system are limited to only the UV emission to ensure that potentially hazardous direct UV irradiation does not occur in the occupied space after installation. The manufacturer cannot control the proper installation by the user or installer. Therefore, any manufacturer requirements and associated irradiance measurements are the only testing that will be expected for a properly installed luminaire. It is critical that upper-room UVGI luminaires adequately emit radiant energy only into the intended irradiation zone (more than 1.8 m above the floor). The purpose of the test protocol in this section is to describe what professionals need to do to ensure safety after installation.

8.1.1 Scope of the Installation Acceptance Testing. The actual risk may depend on the wavelength and intended use; therefore, consideration should be given to the differences between intense UV-C sources and other products. There is a need to consider more than one category of UVGI upper-room luminaires. The required radiometric tests may differ by category (see, for example, **Table 5-2b** or **Table 5-2c** in **Section 5.4**).

8.1.2 Installation Height. Upper-room UV luminaires will have a required and/or a recommended mounting height and position in the room or on a wall. It is very important to follow the lamp system manufacturer's installation instructions. The bottom of the UVGI system should be no lower than 2.1 m above the floor.

8.1.3 Radiometric Measurement Elevation Plane. Where to take measurements is specified in **Table 5.2b**, in **Section 5.4**. The height above the floor at which to perform the below-axis UV-C radiometric measurements for safety is 1.8 m, since the 99th percentile for standing eye height is approximately 1.75 m to 1.8 m (thus, 1.8 m is more conservative) above the floor.

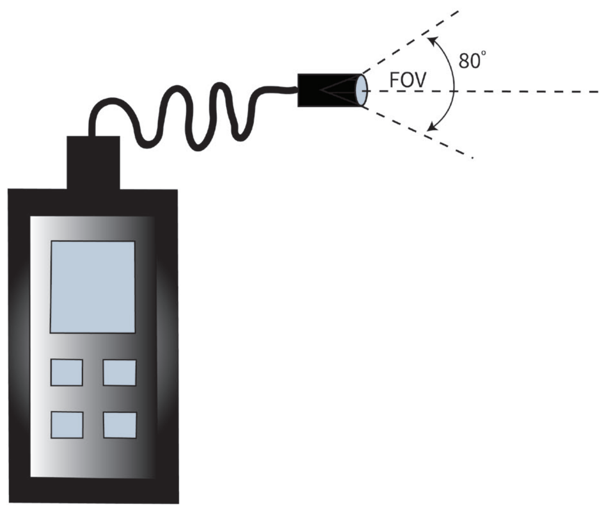
8.1.4 Acceptance Testing.

Test grid location and point spacing:

During installation, testing should be performed on a horizontal plane no higher than 1.8 m above the floor, typically with a grid with 0.5-meter spacing. Alternatively, irradiance simulations of a room can be used to determine areas where the highest irradiance levels are to be expected, and a testing grid can be established in those areas. The testing grid and commissioning plan should be part of the room layout plan provided by the lighting designer.

Detector acceptance angle:

The detector field of view (FOV) and/or orientation should be specified for acceptance testing. The ACGIH/ICNIRP UV exposure guidelines and paragraph 5.1 (Note 3) of IEC 62471:2006[20] specify that the irradiance should be measured with an 80° cone (±40°) for the FOV (see **Figure 8-1**).



Safety meter with 80º cone FOV

Figure 8-1. Instrument for measuring UV-C radiation should have an 80-degree field of view. (Graphic © Illuminating Engineering Society)

Performing the test:
Based on the testing plan grid, the radiometer should be mounted to a holder so that the optical center of the detector will consistently measure at 1.8 m. The initial measurement series should point the radiometer detector approximately horizontally toward the UV lamp system, which will be the worst-case assessment. If multiple systems are installed, measurements should be repeated toward each of the other UV lamp systems. The highest measured irradiance value shall be recorded. No measurement in a room shall exceed the TWA effective irradiation limits based on estimated occupancy exposure duration. (Refer to **Section B.1** for an example of how TWA is applied.)

Since reflection from ceilings cannot always safely be ignored, a test may be performed with the radiometer cone pointing straight upward from selected points in the room, especially where the highest radiation has been measured with the cone pointing horizontally, and close to the lamp system. The FOV shall *not* encompass the lamp system itself. Additional information may be found in *ANSI/IES RP-44-21, Recommended Practice: Ultraviolet Germicidal Irradiation (UVGI)* (see **Section 2.3**).

8.1.5 Adjustable Luminaires. Adjustable luminaires are often favored by competent installers. The four general categories of UV luminaires that allow beam alignment are described in **Section 6.1.2**. These types of adjustable luminaires are designed by manufacturers to aid the installer in making final adjustments to ensure safety and maximal efficacy. Final acceptance testing requires calibrated UV instruments employed by a competent installer.

SIDEBAR: Time-Weighted Averaging (TWA) and Exposure During Installation

Some UV-C exposure may be expected during installation. The important photobiological risk assessment depends upon the cumulative UV-C dose (J/cm2) received by the worker over any eight-hour period. [i] The ACGIH document describes exposure durations for given actinic radiation effective irradiances. An assumption of lengthy (e.g., 8-hour), low irradiance (W/m2), continuous exposure in a standing position, facing the UV-C luminaire, is unreasonable for some UV applications. The ACGIH TLV for UV-C, and other guidelines or standards provided by the manufacturer in the user's manual, should be carefully followed by the installer. Annex B provides an example of the application of TWA.

i. American Conference of Governmental Industrial Hygienists. (2022). 2022. TLVs and BEIs: Based on the Documentation of the Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. Ultraviolet Radiation. Cincinnati, OH: American Conference of Governmental Industrial Hygienists.

8.2 Post-installation Testing of Whole-Room RG-0 and RG-1 UV-C Systems

Use of UV-C luminaires in whole-room systems assumes that people may be present and directly exposed to the UV-C emission. Thus, installed whole-room UV-C systems are assumed to be in RG-0 or RG-1. Whole-room luminaires shall be mounted on the ceiling with their emission directed downward. The concern is not stray radiation from the UV luminaire or reflection from ceiling or walls, as it is for upper-room RG-3 systems, but the actual dose received by a person over a given period.

The information in **Section 8.1** and its subsections on upper-room systems can be applied, with modifications as described in the subsections that follow, to the testing of whole-room UV-C systems.

8.2.1 Scope of the Installation Acceptance Testing. The actual risk or hazard to the skin or eye with a whole-room UV-C system depends strongly on the wavelength of the emitted radiation. Emitted radiation below 240 nm is considered a necessary condition for occupied whole-room UV-C systems. Irradiance below this wavelength provides the dose necessary for multi-log inactivation of viruses in air while simultaneously being low enough to not harm humans exposed for up to an 8-hour day. This occurs because the TLV for wavelengths below 240 nm is more than an order of magnitude higher than that at 254 nm, while the inactivation efficiency of pathogens, particularly viruses, in air is the same as or higher than that due to the emission from low-pressure mercury sources at 254 nm. Commercially available krypton chloride excimer (KrCl*) lamps emit in a band whose peak is at approximately 222 nm. Systems using 254-nm radiation can be designed to be safe but may have unacceptably low inactivation efficacy. Thus, the scope of testing is to ascertain that the irradiance at a given height above the floor is lower than a given maximum, which depends on the given TLV and the TWA exposure duration.

8.2.2 Installation Height. For all whole-room systems installed in occupied rooms, the installation height is the most important parameter to ensure a safe installation. Such systems are installed on the ceiling, and ceiling heights vary considerably. It is very important to follow the lamp system manufacturer's installation instructions. In order to achieve an installation that does not expose individuals above the applicable TWA exposure limits, such a lamp system shall not be mounted at a lower height than specified. Further, if multiple luminaires are installed on the ceiling, the potential increase in the maximum irradiance due to overlap of the UV-C beams needs to be considered. Some whole-room systems contain internal switches to adjust system on-time for different ceiling heights or specific installations.

8.2.3 Elevation Plane for Radiometric Measurements. The procedure here is the same as in **Section 8.1.3**.

8.2.4 Acceptance Testing.

Horizontal-plane irradiance measurements. During installation, testing shall be performed on a horizontal plane no higher than 1.8 m above the floor and at least 1 m from the source. Irradiance values shall be obtained directly below the source(s), with all sources on, and with detector pointed up. For multiple-lamp installations, downward irradiance measurements should also be measured, at a distance of about 1.0 m from the floor, again directly below the source. Such measurements are helpful in determining average values, as the angular irradiance distribution from different manufacturers can vary significantly. Downward irradiance measurements should be done without the 80-degree cone, as these measurements relate to the skin TLVs.

Vertical-plane irradiance measurements. Irradiance measurements of the lamp emission that can enter the eye shall also be made. Vertical-plane measurements shall be made at a height of 1.8 m. Such measurements are best done near a wall where much of the room is in view. The detector field of view (FOV) and/or orientation should be specified for acceptance testing. The ACGIH/ICNIRP UV exposure guidelines and Note 3 in paragraph 5.1 of IEC 62471:2006[20] specify that the irradiance should be measured with an 80° cone (±40°) for the FOV (see **Figure 8-1**).

8.3 User Safety for UV-C Mobile Systems

Safe operation of UV-C mobile systems, including robots and similar equipment in unoccupied rooms is ensured through use of good design and manufacturing processes (see **Section 6**). The professional user shall follow the manufacturer's instructions. Warning labels should be applied on switches and at entry doors to ensure that the space is not occupied during treatment. ANSI/IES RP-44-21 (see **Section 2.3**) provides additional information.

8.4 Instrument Performance Specifications

The performance specifications of the radiometric detector and meter should be provided in any testing report. If a broadband meter is used, it should be fully characterized to provide the $S(\lambda)$-weighted irradiance for the lamp spectrum tested. For example, a low-pressure mercury-vapor lamp emission is dominated by 254 nm, where $S(\lambda)$ is 0.5; hence, the 8-hour TWA irradiance limit of 0.1 $\mu W \cdot cm^{-2}$ effective is actually 0.2 $\mu W \cdot cm^{-2}$ when unweighted.

It is important to note that some meters have both weighted and unweighted scales, while others have only one or the other. The user needs to be fully familiar with the type of instrument being used. The instrument should be sufficiently sensitive to read levels at the 8-hour irradiance limit and at 50% of that limit. The user should follow the manufacturer's instructions with respect to compensating for ambient radiation that is outside of the relevant UV-C range.

Means of calibrating the portable meter used in testing could be provided by the meter manufacturer, based upon spectroradiometric measurements for the type of lamp to be tested, and then spectrally weighted to provide a transfer calibration for the field-test meter. Records of the calibration and measurement results should be recorded.

Annex A – Spectral Weighting – Exposure Limits and Example

The material in this annex is adapted from *ANSI/IES RP-27-20, Recommended Practice: Photobiological Safety for Lighting Systems* (see **Section 2.2**) However, some ACGIH TLVs changed in 2022; the 2022 values are incorporated into this annex and should be applied as described in the subsections that follow.

A.1 General

The UV exposure limits represent conditions under which it is believed that nearly all individuals in the general population may be repeatedly exposed without adverse health effects. However, they do not apply to photosensitive individuals or to individuals concomitantly exposed to photosensitizing agents.[21] Such individuals, in general, are more susceptible to adverse health effects from optical radiation—particularly from UV-A—than individuals who are not photosensitive or concomitantly exposed to photosensitizing agents. The susceptibility of photosensitive individuals varies greatly, and it is not possible to set exposure limits for this portion of the population. The exposure limits provided below should be used as guides in the control of exposure to continuously emitting sources where the exposure duration is 0.1 s or longer. These values should be used as guides in the control of exposure and shall not be regarded as a fine line between safe and dangerous levels.

A.2 Skin and Eye Exposure Limits for 200 nm to 400 nm

The limits for human exposure to UV radiation incident upon the unprotected skin or eye where irradiance values are known and exposure time is controlled are as follows:

- The effective irradiance exposure for the unprotected skin or eye should not exceed the values given in **Table A-1** within an 8-hour period.
- To determine the effective UV irradiance E_S of a broad-band source, the following weighting formula shall be used:

$$E_s = \sum_{200}^{400} E_\lambda \bullet S(\lambda) \bullet \Delta\lambda \,, \qquad \textbf{(A-1a)}$$

where:

E_S = the effective ultraviolet irradiance, W/cm²

E_λ = the spectral irradiance, W/(cm²·nm)

$S(\lambda)$ = the ultraviolet hazard weighting function (see **Table A-2**)

$\Delta\lambda$ = the calculation interval, nm

The summation extends from 200 nm to 400 nm

The permissible exposure time for exposure to UV radiation incident upon the unprotected eye or skin shall be computed by:

$$t_{\max} = (0.003)/E_S \,, \qquad \textbf{(A-1b)}$$

where:

$t_{\max}$ = the permissible exposure time, seconds

E_S = the effective ultraviolet irradiance, W/cm

The value "0.003" is in J/cm²

The exposure time may also be determined using **Table A-1**, which provides exposure times corresponding to an effective irradiance in µW/cm².

Note 1: All the preceding exposure limits for UV radiation apply to sources that subtend an angle of less than 80° (1.4 radians), i.e., sources within 40° of the normal to the irradiance area. Sources that subtend a greater angle need to be measured only over an angle of 80°.

Note 2: A limiting aperture no greater than 2.5-cm diameter shall be used with sources producing a uniform optical radiation pattern. However, with sources of optical radiation that do not produce a uniform optical radiation pattern (i.e., contain hot spots less than 2.5 cm), a 7-mm aperture shall be used. The measurement shall be made in that position of the beam giving the maximum reading.

Table A-1. Permissible Ultraviolet Exposure Duration

Duration of Exposure Per Day		TWA Effective UV Irradiance E_s (μW/cm^2)
8	hours	0.1
4	hours	0.2
2	hours	0.4
1	hour	0.8
30	minutes	1.7
15	minutes	3.3
10	minutes	5
5	minutes	10
1	minute	50
30	seconds	100
10	seconds	300
1	second	3,000
0.5	second	6,000
0.1	second	30,000

The relative spectral weighting function values are shown in **Table A-2** for wavelengths from 200 through 400 nm in 1-nm intervals as published by ACGIH in 2022 for the eye and skin, as well as the former ACGIH spectral effectiveness. The ACGIH TLV values for the eye for radiation between 207 and 240 nm are derived from the formula:

$$\mathrm{TLV}_{\mathrm{eye}} = 100 \cdot 10^{0.067 \cdot (240-\lambda)} \qquad \textbf{(A-2a)}$$

The ACGIH values for the skin for radiation between 200 and 250 nm are derived from the formula:

$$\mathrm{TLV}_{\mathrm{skin}} = 100 \cdot 10^{0.06 \cdot (250-\lambda)} \qquad \textbf{(A-2b)}$$

The relationship between the ACGIH TLV and the spectral effectiveness, S_λ is

$$S_\lambda = 30/TLV \qquad \textbf{(A-3)}$$

A.3 Eye Exposure Limit for 315 nm to 400 nm

For the spectral region 315 nm to 400 nm (within the UV-A), the total UV irradiance for the unprotected eye, E_{UV}, should not exceed 1 mW/cm^2 for periods greater than 1,000 s (approximately 16 minutes). For exposure times less than 1,000 s, the total radiant exposure to the eye should not exceed 1.0 J/cm^2.

Note: All of the preceding exposure levels for UV radiation apply to sources that subtend an angle less than 80°, i.e., sources within 40° of the normal to the irradiance area. Sources that subtend a greater angle need be measured only over an angle of 80°.

This can be expressed as:

$$E_{\mathrm{uv}} \leq 0.001 \quad (\text{for } t \geq 10^3)\ , \qquad \textbf{(A-4a)}$$

$$t_{\max} \leq 1.0/E_{uv} \quad (\text{for } t < 10^3)\ , \qquad \textbf{(A-4b)}$$

where:

t = time, seconds

$t_{\max}$ = time, seconds

E_{uv} = the total ultraviolet irradiance between 315 nm and 400 nm on the unprotected eye in units of W/cm^2; the value 1.0 is in J/cm^2

Notes: (Refer to Notes for **Section A.2**.)

Annex B – Examples: The Use of Time-Weighted Average (TWA) in Risk Group Classification

This annex is not part of ANSI/IES RP-27.1-22, Recommended Practice: Risk Group Classification and Minimizing

Table A-2. Spectral Weighting Function S(λ) in 1-nm Steps[a] for Assessing Actinic Radiation Hazard (Derived From ACGIH TLVs[b])

Wavelength, nm[c]	Current UV Spectral Effectiveness (Eye)	Current UV Spectral Effectiveness (Skin)	Former UV Spectral Effectiveness (Eye and Skin)[d]
200[c]	0.001845	0.000300	0.030000
201	0.001845	0.000344	0.033359
202	0.001845	0.000395	0.037094
203	0.001845	0.000454	0.041247
204	0.001845	0.000521	0.045865
205	0.001845	0.000600	0.051000
206	0.001845	0.000689	0.055089
207	0.001846	0.000791	0.059507
208	0.002153	0.000908	0.064278
209	0.002513	0.001042	0.069433
210	0.002930	0.001196	0.075000
211	0.003419	0.001374	0.078631
212	0.003989	0.001577	0.082438
213	0.004655	0.001810	0.086429
214	0.005434	0.002078	0.090613
215	0.006340	0.002386	0.095000
216	0.007398	0.002739	0.099544
217	0.008632	0.003144	0.104305
218	0.010071	0.003610	0.109294
219	0.011751	0.004144	0.114522
220	0.013712	0.004758	0.120000
221	0.015999	0.005462	0.125477
222	0.018668	0.006270	0.131203
223	0.021782	0.007199	0.137192
224	0.025415	0.008264	0.143453
225	0.029654	0.009487	0.150000
226	0.034601	0.010892	0.157262
227	0.040373	0.012504	0.164876
228	0.047107	0.014355	0.172858
229	0.054965	0.016480	0.181226
230	0.064134	0.018919	0.190000
231	0.074832	0.021720	0.199088
232	0.087315	0.024935	0.208611
233	0.101880	0.028625	0.218589
234	0.118875	0.032863	0.229044
235	0.138704	0.037727	0.240000
236	0.161841	0.043311	0.250953

237	0.188838	0.049722	0.262407
238	0.220338	0.057082	0.274383
239	0.257092	0.065532	0.286906
240	0.300000	0.075232	0.300000
241	0.311141	0.086368	0.311141
242	0.322696	0.099152	0.322696
243	0.334680	0.113829	0.334680
244	0.347109	0.130678	0.347109
245	0.360000	0.150021	0.360000
246	0.373023	0.172227	0.373023
247	0.386517	0.197720	0.386517
248	0.400500	0.226987	0.400500
249	0.414988	0.260586	0.414988
250	0.430000	0.300000	0.430000
251	0.446523	0.300000	0.446523
252	0.463681	0.300000	0.463681
253	0.481498	0.300000	0.481498
254	0.500000	0.300000	0.500000
255	0.520000	0.300000	0.520000
256	0.543733	0.300000	0.543733
257	0.568548	0.300000	0.568548
258	0.594497	0.300000	0.594497
259	0.621629	0.300000	0.621629
260	0.650000	0.300000	0.650000
261	0.679247	0.300000	0.679247
262	0.709810	0.300000	0.709810
263	0.741748	0.300000	0.741748
264	0.775123	0.300000	0.775123
265	0.810000	0.300000	0.810000
266	0.844866	0.300000	0.844866
267	0.881234	0.300000	0.881234
268	0.919166	0.300000	0.919166
269	0.958732	0.300000	0.958732
270	1.000000	0.300000	1.000000
271	0.991869	0.300000	0.991869
272	0.983804	0.300000	0.983804
273	0.975804	0.300000	0.975804
274	0.967870	0.300000	0.967870
275	0.960000	0.300000	0.960000
276	0.943438	0.300000	0.943438

277	0.927162	0.300000	0.927162
278	0.911167	0.300000	0.911167
279	0.895448	0.300000	0.895448
280	0.880000	0.300000	0.880000
281	0.856810	0.300000	0.856810
282	0.834230	0.300000	0.834230
283	0.812246	0.300000	0.812246
284	0.790841	0.300000	0.790841
285	0.770000	0.300000	0.770000
286	0.742042	0.300000	0.742042
287	0.715099	0.300000	0.715099
288	0.689135	0.300000	0.689135
289	0.664113	0.300000	0.664113
290	0.640000	0.300000	0.640000
291	0.618618	0.300000	0.618618
292	0.597951	0.300000	0.597951
293	0.577974	0.300000	0.577974
294	0.558664	0.300000	0.558664
295	0.540000	0.300000	0.540000
296	0.498397	0.300000	0.498397
297	0.460000	0.300000	0.460000
298	0.398914	0.300000	0.398914
299	0.345939	0.300000	0.345939
300	0.300000	0.300000	0.300000
301	0.221042	0.221042	0.221042
302	0.162865	0.162865	0.162865
303	0.120000	0.120000	0.120000
304	0.084853	0.084853	0.084853
305	0.060000	0.060000	0.060000
306	0.045404	0.045404	0.045404
307	0.034358	0.034358	0.034358
308	0.026000	0.026000	0.026000
309	0.019748	0.019748	0.019748
310	0.015000	0.015000	0.015000
311	0.011052	0.011052	0.011052
312	0.008143	0.008143	0.008143
313	0.006000	0.006000	0.006000
314	0.004243	0.004243	0.004243
315	0.003000	0.003000	0.003000
316	0.002400	0.002400	0.002400

317	0.002000	0.002000	0.002000
318	0.001600	0.001600	0.001600
319	0.001200	0.001200	0.001200
320	0.001000	0.001000	0.001000
321	0.000819	0.000819	0.000819
322	0.000670	0.000670	0.000670
323	0.000540	0.000540	0.000540
324	0.000520	0.000520	0.000520
325	0.000500	0.000500	0.000500
326	0.000479	0.000479	0.000479
327	0.000459	0.000459	0.000459
328	0.000440	0.000440	0.000440
329	0.000425	0.000425	0.000425
330	0.000410	0.000410	0.000410
331	0.000396	0.000396	0.000396
332	0.000383	0.000383	0.000383
333	0.000370	0.000370	0.000370
334	0.000355	0.000355	0.000355
335	0.000340	0.000340	0.000340
336	0.000327	0.000327	0.000327
337	0.000315	0.000315	0.000315
338	0.000303	0.000303	0.000303
339	0.000291	0.000291	0.000291
340	0.000280	0.000280	0.000280
341	0.000271	0.000271	0.000271
342	0.000263	0.000263	0.000263
343	0.000255	0.000255	0.000255
344	0.000248	0.000248	0.000248
345	0.000240	0.000240	0.000240
346	0.000231	0.000231	0.000231
347	0.000223	0.000223	0.000223
348	0.000215	0.000215	0.000215
349	0.000207	0.000207	0.000207
350	0.000200	0.000200	0.000200
351	0.000191	0.000191	0.000191
352	0.000183	0.000183	0.000183
353	0.000175	0.000175	0.000175
354	0.000167	0.000167	0.000167
355	0.000160	0.000160	0.000160
356	0.000153	0.000153	0.000153

357	0.000147	0.000147	0.000147
358	0.000141	0.000141	0.000141
359	0.000136	0.000136	0.000136
360	0.000130	0.000130	0.000130
361	0.000126	0.000126	0.000126
362	0.000122	0.000122	0.000122
363	0.000118	0.000118	0.000118
364	0.000114	0.000114	0.000114
365	0.000110	0.000110	0.000110
366	0.000106	0.000106	0.000106
367	0.000103	0.000103	0.000103
368	0.000099	0.000099	0.000099
369	0.000096	0.000096	0.000096
370	0.000093	0.000093	0.000093
371	0.000090	0.000090	0.000090
372	0.000086	0.000086	0.000086
373	0.000083	0.000083	0.000083
374	0.000080	0.000080	0.000080
375	0.000077	0.000077	0.000077
376	0.000074	0.000074	0.000074
377	0.000072	0.000072	0.000072
378	0.000069	0.000069	0.000069
379	0.000066	0.000066	0.000066
380	0.000064	0.000064	0.000064
381	0.000062	0.000062	0.000062
382	0.000059	0.000059	0.000059
383	0.000057	0.000057	0.000057
384	0.000055	0.000055	0.000055
385	0.000053	0.000053	0.000053
386	0.000051	0.000051	0.000051
387	0.000049	0.000049	0.000049
388	0.000047	0.000047	0.000047
389	0.000046	0.000046	0.000046
390	0.000044	0.000044	0.000044
391	0.000042	0.000042	0.000042
392	0.000041	0.000041	0.000041
393	0.000039	0.000039	0.000039
394	0.000037	0.000037	0.000037
395	0.000036	0.000036	0.000036
396	0.000035	0.000035	0.000035

397	0.000033	0.000033	0.000033
398	0.000032	0.000032	0.000032
399	0.000031	0.000031	0.000031
400	0.000030	0.000030	0.000030

Table notes:

a. The 1-nm interval values between the ACGIH-published 5-nm-interval values for the eye for radiation below 240 nm and for the skin for radiation below 250 nm were derived by application of **Equations A-2a** or **A-2b** and **A-3**.

b. The values for $S(\lambda)$ were derived from ACGIH TLVs for UV radiation (adopted January 2022).[4] These values are used in **Equation A-1a** for determining actinic UV hazards.

c. The $S(\lambda)$ value for wavelengths from 180 to 200 nm is 0.00184 for the eye and 0.0003 for the skin.

d. Values shown for historical reference only; they are not to be used.

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Time-weighted averaging of an exposure to achieve a TWA irradiance for applications results in the assessment distances shown in **Tables 5-2a** through **5-2c** (see **Section 5.4**). In addition, the user may need to employ TWA for assessing risks from RG-2 or RG-3 lamp systems, such as luminaires employed in upper-room or whole-room UVGI applications. Two examples are provided in this annex to show how the concept of TWA is applied in risk group classification.

From **Equation A-1b** in **Annex A** it can be seen that the permissible time for exposure of unprotected eyes or skin to UV radiation shall be computed from:

$$t_{\mathrm{TWA}} = (0.003)/E_{\mathrm{S}}\ , \qquad \textbf{(B-1)}$$

where:

t_{TWA} = the permissible exposure time in seconds, applying the TWA concept

E_{S} = the effective ultraviolet irradiance in W/cm^2.

The units for "0.003" in the equation are J/cm^2

Rearranging **Equation B-1** gives maximum effective irradiance:

$$E_{\mathrm{S}} = (0.003)/t_{\mathrm{TWA}} \qquad \textbf{(B-2)}$$

In the case of exposures for 8 hours (rounded up to 30,000 s) in real time, if there are no constraints, then $E_s = 0.1$ µW/cm^2.

However, relevant factors such as source wavelength, location of the eyes in a room, and motion of the person within the room, including leaving the room for a time, can all increase the effective irradiance that is allowed. It may also be surmised that the irradiance of the eye is a major issue in determining the risk. Limits were derived based on protecting the eye, whereas these limits are only starting points for the far less-sensitive skin tissues.

B.1 Example of Time Weighting of Exposure in a Space with Upper-Room GUV

A good example of how time-weighted averaging (TWA) is employed to assess risk from scattered UV-C in the occupied space is exemplified by an upper-room GUV installation with a UV luminaire employing either a traditional low-pressure mercury lamp (see **Figure B-1**) or LEDs emitting in the UV-C region.

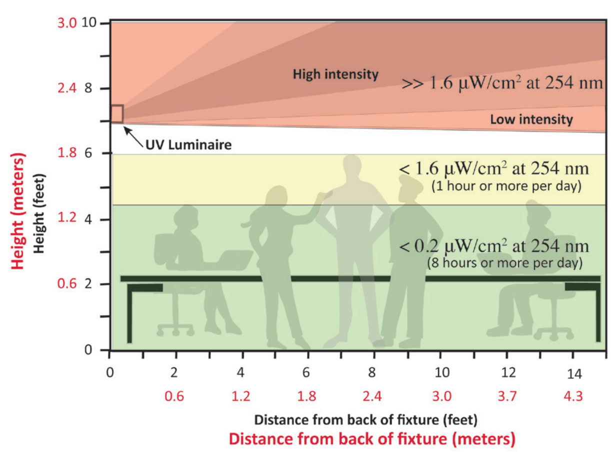

Figure B-1. Example of how an occupational hygienist might determine different zones of exposure by time-weighting. The acceptable values in the white zone will vary depending on ceiling height. (Graphic © Illuminating Engineering Society)

In this example, taken from *ANSI/IES RP-44-21, Recommended Practice: Ultraviolet Germicidal Irradiance (UVGI)* (see **Section 2.3**), a time-motion evaluation provided a maximal standing duration of 1 hour in one day, so a 1-hour TWA at an eye level of 1.8 m was assessed. In addition, there were seated positions (eye level 1.3 m above the floor) that required an 8-hour (RG-0) limit for the *effective* irradiance of 0.2 μW/cm². It should be noted that the value of 0.2 μW/cm² is derived from **Table A-2** in **Annex A** and shows that the effectiveness of irradiance at 254 nm is 0.5.[22]

In a given work situation using upper-room UVGI, e.g., a hospital or school, the cumulative exposure can be obtained through use of UV-C-sensitive badges or other devices that accumulate the exposure. An example is shown in **Figure B-2**.

B.2 Example of Time-Weighting of Irradiance for Whole-Room Air Disinfection Using KrCl* Lamps

Excimer lamps are being developed for germicidal use, particularly for whole-room air disinfection in occupied spaces. The krypton-chloride excimer (KrCl*) lamp's emission peak is at about 222 nm, and almost all of the radiated power from the lamp is emitted within a few nanometers on each side of that peak. It should be noted that radiation in this wavelength region has been found very effective at inactivating viruses such as SARS-CoV-2 (responsible for COVID-19).[23,24,25,26,27]

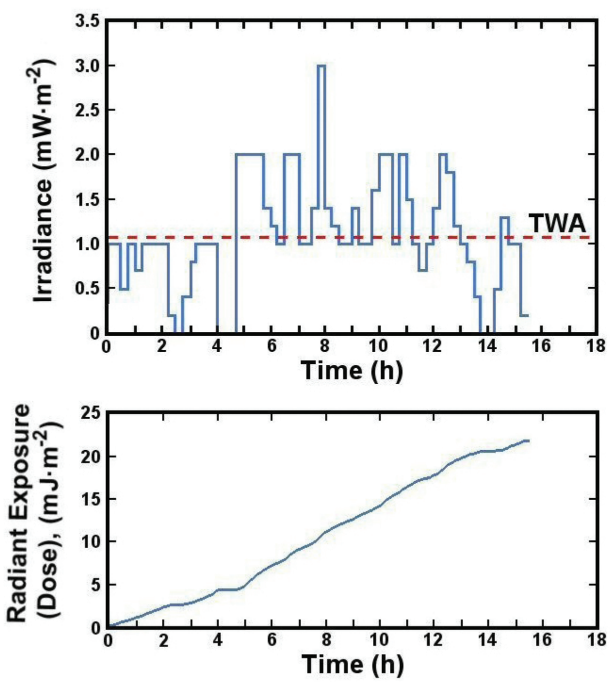

Figure B-2. Time-weighted averaging (TWA). ***Top:*** **The effective irradiance pattern during a day for an individual in a UVGI setting.** ***Bottom:*** **The accumulated effective radiant exposure (photobiological "dose") over the same time period.**

One concern with the KrCl* lamp is emission at 259 nm, and in a broad band in the UV-B region from 280 to 300 nm that has become important for evaluating the hazard to eyes and skin when the 222-nm emission is at or near the TLV. Even though these peaks contain less than 5% of the total power radiated, the penetration of the longer-wavelength emission through the stratum corneum and into the basal (germinative) layer of the epidermis creates a potential hazard for delayed skin effects. Longer UV wavelengths also penetrate the tear layer and pose a potential eye hazard.[28,29,30] Thus, it is recommended that KrCl* lamps or luminaires be filtered to minimize risk.

The current ACGIH TLVs for the UV show the spectral weighting factor, S(λ), at 222 nm for the eye and skin as 0.018668 and 0.006270, respectively. Studies[25,29] have shown that both the skin and eye are significantly less sensitive to radiation at 222 nm than that implied by the previous ACGIH spectral sensitivity curve (2021 and earlier). For this reason, in 2022 ACGIH increased the limits below 250 nm (see **Table A-2** in **Annex A**), with separate values for the eye and skin.[4,30]

For someone working in a room for 8 hours where excimer lamps are being employed continuously to disinfect the air in the room, i.e., mounted on the ceiling, the maximum irradiance in occupied spaces below 1.8 m (see **Figure B-1**) would be E_s = 5.4 μW/cm² for the eye and 16 μW/cm² for the skin (spectrally weighted for a krypton chloride lamp with an emission centered at 222 nm, and then integrated between approximately 210 and 230 nm). The latter irradiance value is more than 20 times higher than that for low-pressure mercury sources.

The eye is normally shielded by the upper lid from optical radiation coming from above (e.g., the ceiling).

In this case the upper eyelid shields the eye from UV emission that is more than about 40 degrees above the horizontal plane. Further, since the 222-nm band emission from current KrCl* sources sold for whole-room disinfection are more like reflector flood lamps, there is almost no emission from the source that reaches the eye below 40 degrees above horizontal. Therefore, one expects that eye exposure from the source would be significantly less than the exposure for the top of the head.[31] Thus, it can be argued that the skin TLV limit is the appropriate measure, i.e., 16 μW/cm^2 over an 8-hour day, or 127 μW/cm^2 in one hour.[4] This irradiance level is large enough to continuously inactivate many viruses in room air, with a one-log reduction in durations on the order of a few minutes. In the case of fixtures mounted in the ceiling and pointed downward, such inactivation rates are very difficult to reach with air exchange methods.

Annex C – Biological Effects of Ultraviolet Radiation

This annex is not part of ANSI/IES RP-27.1-22, Recommended Practice: Risk Group Classification and Minimizing Photobiological Hazards from Ultraviolet Lamps and Lamp Systems. *It is provided for informative purposes only.*

C.1 Effects on Skin

C.1.1 Sunburn. Minor sunburn is a skin reddening (actinic erythema) that appears up to 12 hours after exposure to UV radiation. Erythema gradually fades after a few days, replaced by some tanning in individuals with tanning capability. Severe sunburn is painful and results in inflammation, blistering, and peeling of the skin. Aside from photoimmunological effects (refer to **Section C.2**), systemic effects of severe sunburn are unknown except for transient fever. Sunburn severity depends critically on skin phototype (see **Table C-1**) and UV dose. For fair-skinned people (skin phototypes I and II, melano-compromised), the relative effectiveness of UV radiation for tanning and for erythema is approximately the same over the entire range of UV-B and UV-A wavelengths.[32] For people who tan well and who rarely sunburn (skin phototypes III and IV, melano-competent), the tanning efficacy of UV-A radiation is higher than its erythemal efficacy.[33] There is a range of radiant exposures (expressed in terms of standard erythemal dose, SED, which is used to quantify an individual minimal erythemal dose, MED [see **Sidebar**]) that will produce noticeable effects in people of different skin phototypes. UV-C radiation generally produces only a very mild erythema that is less long-lasting than longer wavelengths produce, because of its very shallow penetration.

SIDEBAR – Minimum Erythemal Dose

A minimum erythemal dose (MED) is defined as the UV exposure that will produce a “just noticeable” erythema on previously unexposed skin of an individual. If the exposure is spectrally weighted by the CIE erythemal action spectrum, the MED corresponds to an effective radiant exposure expressed in standard erythemal dose (SED) units, depending on individual skin phototype (see **Table C-1, Section C.1.5**). Erythemally effective irradiances (W·m^{-2} eff.), expressed in SEDs per hour, can be calculated for any source of UV radiation using spectral irradiance data for the source, at the point of interest, and the relative spectral weighting factors for erythema, promulgated by the International Commission on Illumination (CIE).[i,ii] One SED is an ISO/CIE official unit,[ii] and an effective radiant exposure of 100 J·m^{-2}.

i. McKinlay AF, Diffey BL. A reference action spectrum for ultraviolet induced erythema in human skin. CIEJ. 1987;6(1):17-22.

ii International Standards Organization (ISO) and International Commission on Illumination (CIE). Erythemal Reference Action Spectrum and Standard Erythema Dose. (ISO 17166:1999; CIE S007/E-1998). Vienna, CIE; 1999.

C.1.2 Skin Cancer. The most serious long-term effects attributed to exposure of the skin to UV radiation exposure are skin cancers.[34,35,36] Squamous cell and basal cell carcinomas are common, rarely fatal forms of skin cancer, which are often referred to collectively as non-melanoma skin cancers (NMSC). Experimental studies clearly indicate that UV-A and UV-B radiation can cause squamous cell carcinomas (SCC) in mice. The same effects are likely in humans.[37,38] The erythemal potential of a UV source is gaining acceptance as a reasonable approximation for a quantitative measure of its carcinogenic potential in the UV-B range.[39] A series of publications[40,41,42,43] served as a source for CIE/ISO Standard 28077:2016, which provides a standardized action spectrum for photocarcinogenesis. For individuals who insist on acquiring a tan despite the health risks, there are conclusions that can be drawn from the scientific data to minimize risk.[44]

Cutaneous malignant melanoma (MM), while much less frequent than NMSC, is much more serious and accounts for the majority of deaths from skin cancer. There are some mammalian data[45,46] for MM that indicate a strong UV-B-melanoma etiology, but these data are not entirely consistent with data from a fish melanoma model that indicate a strong implication of UV-A radiation in addition to UV-B radiation in the induction of fish melanoma.[47] The animal models demonstrate an increased effect of neonatal exposure on tumor induction, including malignancy.[48,49] The evidence for the indictment of sunlight as a causal agent is limited to epidemiological data. The data indicate that intermittent exposure to high levels of solar UV radiation, particularly at an early age, may be a contributing causal factor (relevant reviews are available[50,51]). The individual risk of MM is higher in people who have a large number of nevi (moles) and who sunburn readily and tan poorly with exposure to solar UV radiation. Some data suggesting an association between the use of sunbeds and an increased risk of MM have been published, but the relative importance of UV-B and UV-A radiation and other factors (especially general sun-seeking behavior) in causing this association is unclear.[52,53,54,55,56,57,58,59,60]

Taken collectively, both experimental and epidemiological data indicate that cumulative exposure increases the risk for skin cancers. This includes childhood exposures. The action spectrum for NMSC is centered in the UV-B region and is greatly reduced in both the UV-C and UV-A spectral regions.[34,61] There is no evidence at this time of carcinogenic risk in humans from UV-C exposure.

C.1.3 Premature Skin Aging. There is considerable evidence that cumulative UV-A and UV-B exposure results in premature skin aging characterized by a dry, coarse, leathery, and wrinkled appearance. It has been clearly demonstrated that UV-A radiation causes skin damage in mice.[62,63,64]

Daily exposures to sub-erythemogenic purely UV-A radiation within the spectral region 320 to 400 nm for eight days, or exposure to longer UV-A wavelengths between 340 and 400 nm for three months, results in cumulative morphological skin alterations, indicative of tissue injury.[65,66,67] In a 5-year longitudinal study of women who used or did not use tanning salons,[68] serious modifications of skin elasticity and extensibility were found in a tanning-salon-user group.

C.1.4 Photodermatosis and Photosensitivity. Polymorphic light eruption (PLE) is a common photodermatosis readily produced in some people by exposure to UV sunbed radiation.[69] Other photo-aggravated dermatoses such as systemic lupus erythematosus are also exacerbated by the use of UV sunbeds.[70] Certain medicines and chemical and topical products, such as perfumes and lotions, may cause skin photosensitization in sunbed users.

C.1.5 Tanning. When skin is exposed to ultraviolet radiation, two distinct tanning reactions ensue:

- **Immediate pigment darkening (IPD).** IPD begins immediately on exposure to UV radiation and is caused by the darkening of the pigment melanin, already present in the skin; it is normally seen only in people who have at least a moderate constitutive tan. Such pigmentation begins to fade within a few minutes after cessation of exposure. Radiation between 320 and 400 nm is regarded as being most effective for IPD.[71]
- **Delayed tanning (neo-melanogenesis).** The visible pigmentation takes at least 3 days to develop and results from exposure to UV-A radiation but is more effectively produced by UV-B radiation.[32,33] UV radiation-induced neo-melanogenesis is strongly dependent on cellular responses arising from UV radiation-induced DNA damage in cellular nuclei.[72,73] Delayed tanning is more persistent than IPD and results from an increase in the number, size,

and pigmentation of melanin granules.[74] Exposure to UV-B radiation results also in an increase in the thickness and scattering properties of the epidermis (the outer layer of the skin).[62,75] The degree to which an individual successfully tans as the result of UV radiation depends critically on his or her skin phototype. These phototypes (classified as phototypes I through VI) are presented in **Table C-1**. In principle, the tanning or sun-burning reaction of a person to UV-radiation is similar, whether the exposure is from an artificial UV source or from solar radiation.

C.2 Photoimmunological Effects

Exposure to UV radiation causes localized skin and systemic modifications due to photoimmunological reactions.[77] also a particular concern with UV-A sunbed radiation.[69,78] The clinical improvement of atopic dermatitis with sub-erythemal UV-A exposure indicates that UV radiation is able to modify both substantially normal and pathological immunological reactions. UV-B radiation can promote the development of skin cancer, perhaps by suppressing the immune system, thus allowing the tumor to escape immune surveillance.[79] The action spectra in the UV-B region for mixed lymphocyte reaction and mixed epidermal cell lymphocyte reaction were found to be similar to the induction of thymine dimers upon DNA irradiation.[80]

There is also evidence that exposure to UV radiation can activate and accelerate the growth of viruses afflicting humans,[81,82] including human immunodeficiency virus (HIV),[83] and have effects on infectious disease.[84] At present, the significance of these observations with respect to human health is unclear. The effects of UV-A exposure on the immune system are even more uncertain.[85,86]

C.3 Ocular Effects

C.3.1 The Cornea. The principal adverse effect of the absorption of UV-C and UV-B radiation by the cornea is termed photokeratitis (sometimes called “welder’s flash” or “snow-blindness”), and damage is generally limited to the epithelial (front surface) cells of the cornea.[87] After a 6-to-12-hour latent period that depends inversely on the severity of the exposure, there is severe corneal pain, photophobia, lacrimation, and eyelid spasm. These symptoms typically resolve in 24 hours. Photokeratitis was reported as a result of consumer use of some unsafe germicidal products during the COVID-19 pandemic.[88] There is some evidence of possible long-term effects of absorption by the cornea of UV radiation[89] and evidence of endothelial thinning.[90] UV-B photokeratitis thresholds vary greatly by wavelength and have been re-evaluated based on the spectral bandwidth employed in threshold studies.[91] UV-C photokeratitis thresholds have more recently been published.[29]

C.3.2 The Lens. Transmission of UV-C to the lens of the eye is blocked by the cornea, and transmission of UV-A radiation to the lens is much greater than that of UV-B. Animal data indicate that threshold lenticular damage is limited mainly to UV exposure in the wavelength band 295 to 325 nm. Experimental spectral efficiency

Table C-1. Classification of Skin Phototypes Based on Susceptibility to Sunburn in Sunlight and Ability to Tan

Skin Phototype	Sun Sensitivity	Sunburn Susceptibility[a]	Tanning Ability	Classes of Individuals
I	Very sensitive	Always sunburn (<2 SED[b])	No tan	Melano-compromized[c]
II	Moderately sensitive	High (2 – 3 SED)	Light tan	Melano-compromized[c]
III	Moderately insensitive	Moderate (3 - 5 SED)	Medium tan	Melano-competent
IV	Moderately resistant	Low (5 – 7 SED)	Dark tan	Melano-competent
V	Resistant	Very low (7 – 10 SED)	Natural brown skin	Melano-protected
VI	Very resistant	Extremely low (>10 SED)	Natural black skin	Melano-protected

Table notes:

a. The ranges of SEDs in parentheses are approximate skin phototypes.
b. SED = standard erythemal dose; see **Section C.1.2**.
c. Melano-compromised individuals have a far greater risk of developing skin cancers than melano-competent individuals.

Source: Fitzpatrick et al., 1995.[76]

for acute cataracts in laboratory animals has been measured only in this spectral region. Aging of the lens is characterized in part by loss of elasticity and browning (brunescence), both of which may be caused partly by UV-A radiation. Most energy absorbed by the lens is re-emitted as longer-wavelength fluorescence.[92] Certain medicines may act as UV-A photosensitizers of the lens.[93]

C.3.3 The Retina. At least for the chronic effects of exposure to solar radiation and to radiation from conventional light sources, the most important retinal-damage mechanism is photochemical injury from short-wavelength light.[87] The gradual brunescence of the lens as it ages results in its decreased transmission of blue light and of UV radiation, thereby affording increased protection to the retina. Young children and people who have had a lens surgically removed (aphakes) are at a higher risk of retinal damage from UV radiation and blue light. Until the 1990s, many implanted artificial lenses did not effectively absorb UV-A radiation.

The lens and cornea provide considerable protection to the retina from most UV radiation, even without protective goggles. The lens blocks UV radiation below 400 nm, and the cornea blocks below 300 nm, but trace amounts of UV-B radiation between 300 and 315 nm may reach the retina, particularly in very young children.[94,95,96]

C.3.4 Protective Eyewear. The use of protective eyewear (goggles) will prevent exposure of the eyes to harmful levels of UV radiation. This represents a very important issue for exposures to artificial sources along one's line-of-sight, since an individual is normally protected from most of the overhead solar UV radiation due to shading by the brow ridge and upper lids. Furthermore, oblique rays can be focused into the nasal equatorial region of the lens.[97,98] UV radiation can only reach the critically important germinative region of the lens by the focusing of oblique rays. Eyewear that does not incorporate side protection is not suitable to protect the eye from lateral UV radiation.

ADDITIONAL READING

In addition to the many items found in the **References** section, the documents cited below may be helpful or informative.

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: ______________________________
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2. Title of publications and year published ______________________________
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4. My proposal (check one):
   [ ] Change to read as follows
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Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____
____
____

6. Reason and substantiation:

____
____
____

Select as applicable:
   [ ] Additional pages are attached. Number of additional pages:
   [ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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